

ORANGE CITY, IA, USA

WMO: 725489

Lat: 42.990N

Lon: 96.063W

Elev: 431

StdP: 96.25

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.8	-26.3	0.4	-21.6	-23.7	0.5	-18.6	14.5	-6.6	12.5	-5.4	3.1	300	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.9	23.9	32.1	23.4	30.1	22.6	26.7	30.2	25.4	29.3	24.2	28.5	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	22.4	29.1	24.0	19.9	28.1	22.7	18.3	27.1	86.4	30.2	80.4	29.5	75.5	28.4	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.6	9.0	DB	-26.2	35.6	3.1	1.8	-28.4	36.8	-30.2	37.9	-31.9	38.8	-34.2	40.1	(4)
(5)			WB	-26.0	28.1	3.4	1.6	-28.5	29.3	-30.5	30.2	-32.4	31.1	-34.8	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-6.7	-5.0	2.3	9.5	15.9	21.6	23.6	22.2	18.2	10.6	2.6	-5.0	(6)	
	DBStd	12.13	7.02	6.82	6.96	5.66	5.01	3.77	3.31	3.01	4.70	5.38	5.98	6.54	(7)	
	HDD10.0	2032	519	421	252	76	8	0	0	0	2	58	231	465	(8)	
	HDD18.3	3830	777	654	496	269	108	14	3	5	61	247	472	723	(9)	
	CDD10.0	1743	0	0	15	61	192	347	422	377	247	76	8	0	(10)	
	CDD18.3	499	0	0	1	3	33	111	166	124	56	6	0	0	(11)	
	CDH23.3	4600	0	0	6	63	349	1043	1596	981	492	68	2	0	(12)	
	CDH26.7	1567	0	0	1	16	116	371	596	297	154	16	0	0	(13)	
(14) Wind	WSAvg	4.0	4.2	4.2	4.5	4.8	4.5	3.9	3.0	2.8	3.4	3.9	4.2	4.1	(14)	
(15) Precipitation	PrecAvg	733	17	17	43	76	90	112	95	89	83	58	31	22	(15)	
	PrecMax	973	33	30	111	126	180	231	234	211	227	143	94	58	(16)	
	PrecMin	563	3	1	0	17	27	29	5	22	21	15	0	1	(17)	
	PrecStd	108	9	7	28	28	38	45	55	47	46	41	25	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.6	23.8	28.8	33.0	34.2	35.0	33.7	32.9	28.9	21.2	12.1	(19)	
		MCWB	6.4	8.1	13.7	15.8	20.0	23.1	25.5	23.8	21.4	17.6	13.4	8.6	(20)	
	2%	DB	7.0	8.0	19.9	24.9	29.5	32.4	32.9	32.0	30.2	25.0	17.3	7.7	(21)	
		MCWB	3.8	5.2	12.8	14.8	19.0	22.5	25.3	24.3	21.2	15.8	11.5	4.5	(22)	
	5%	DB	3.7	5.9	16.2	22.1	27.3	30.9	32.1	29.8	27.9	22.3	13.7	5.0	(23)	
		MCWB	2.0	3.7	11.6	13.4	17.3	21.8	24.9	24.2	19.9	14.7	8.7	2.8	(24)	
	10%	DB	2.3	3.6	12.5	18.9	24.8	28.8	30.0	27.8	26.3	19.1	11.1	2.8	(25)	
		MCWB	1.1	2.0	8.6	11.9	16.3	20.8	24.0	22.4	19.0	13.7	7.4	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	8.4	17.2	18.0	23.6	26.0	29.1	27.9	24.7	21.4	15.2	9.4	(27)	
		MCDB	10.6	11.3	19.8	24.5	28.5	31.3	32.6	30.4	29.1	25.2	19.8	11.2	(28)	
	2%	WB	3.8	5.7	14.2	16.1	20.6	24.6	27.3	26.3	23.4	17.9	11.9	5.0	(29)	
		MCDB	6.0	8.2	17.3	22.6	26.7	29.7	30.7	29.7	27.6	22.1	16.1	7.6	(30)	
	5%	WB	2.3	3.7	11.6	14.3	19.1	23.4	25.8	25.0	22.1	16.0	9.6	3.1	(31)	
		MCDB	3.7	5.8	15.9	20.1	24.8	28.6	29.6	28.3	26.5	20.5	13.1	4.7	(32)	
	10%	WB	0.9	2.1	8.6	12.7	17.6	22.0	24.7	23.6	20.7	13.8	7.3	1.8	(33)	
		MCDB	2.0	3.6	12.8	18.0	22.4	27.0	28.9	27.2	24.6	18.2	10.1	3.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.0	10.1	11.6	11.1	10.6	10.6	10.2	11.7	11.1	9.9	8.1	(35)	
		MCDBR	11.5	11.1	16.1	16.4	14.5	12.4	11.5	11.9	13.7	14.8	14.2	10.8	(36)	
	5% WB	MCWBR	9.1	8.5	9.7	8.4	6.7	5.7	6.2	6.6	6.7	7.8	8.8	8.4	(37)	
		MCWBR	10.5	10.4	14.5	14.1	11.9	10.5	10.1	10.3	11.2	12.4	13.1	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.273	0.284	0.318	0.369	0.393	0.400	0.410	0.408	0.362	0.328	0.291	0.277	(40)	
		taud	2.431	2.421	2.412	2.292	2.297	2.335	2.300	2.293	2.395	2.468	2.491	2.443	(41)	
	Edn at Noon	Edn at Noon	876	924	927	897	881	872	858	849	867	858	846	835	(42)	
		Edh at Noon	74	89	103	125	128	124	127	124	104	85	70	67	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.91	2.99	3.84	4.84	5.63	6.29	6.61	5.51	4.43	3.03	2.09	1.55	(44)	
	RadStd	RadStd	0.17	0.24	0.31	0.41	0.41	0.45	0.30	0.40	0.30	0.37	0.13	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page